
Lab title:- Network Scanning with Nmap

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Tools Used

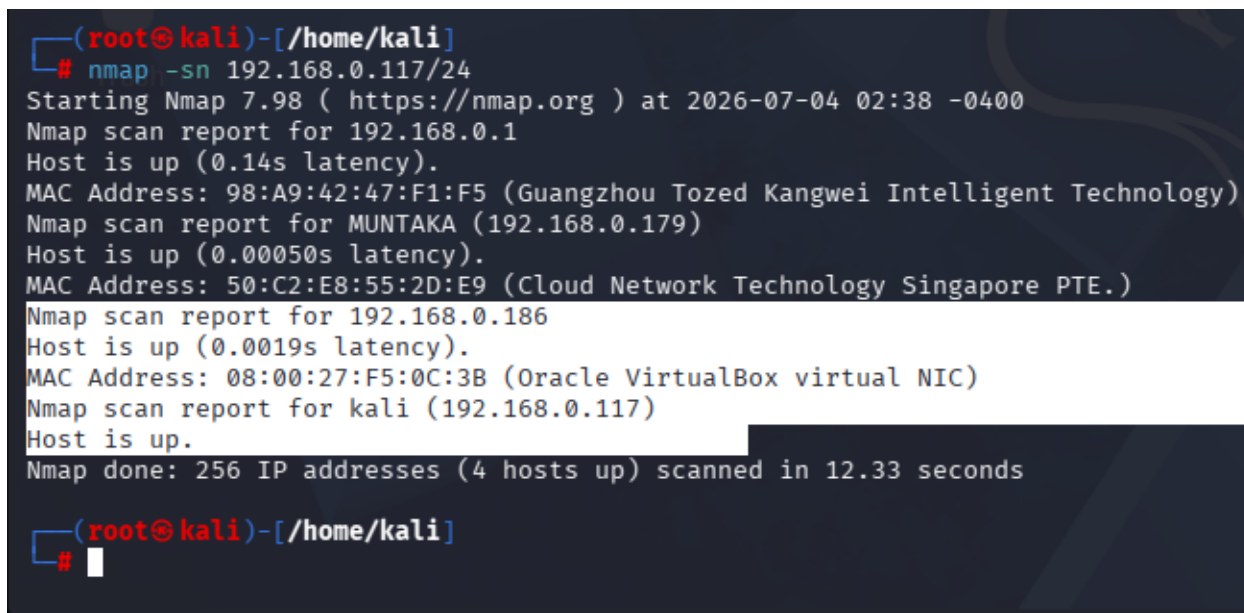
- Kali Linux
 - Nmap
 - VMware Workstation
 - Metasploitable
-

Phase 1: Host Discovery

Command:- nmap -sn 192.168.0.117/24

Purpose :-This scan identifies active hosts on the network without performing a port scan.

Screenshot



```
(root@kali)-[/home/kali]
# nmap -sn 192.168.0.117/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-04 02:38 -0400
Nmap scan report for 192.168.0.1
Host is up (0.14s latency).
MAC Address: 98:A9:42:47:F1:F5 (Guangzhou Tozed Kangwei Intelligent Technology)
Nmap scan report for MUNTAKA (192.168.0.179)
Host is up (0.00050s latency).
MAC Address: 50:C2:E8:55:2D:E9 (Cloud Network Technology Singapore PTE.)
Nmap scan report for 192.168.0.186
Host is up (0.0019s latency).
MAC Address: 08:00:27:F5:0C:3B (Oracle VirtualBox virtual NIC)
Nmap scan report for kali (192.168.0.117)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 12.33 seconds

(root@kali)-[/home/kali]
#
```

Findings

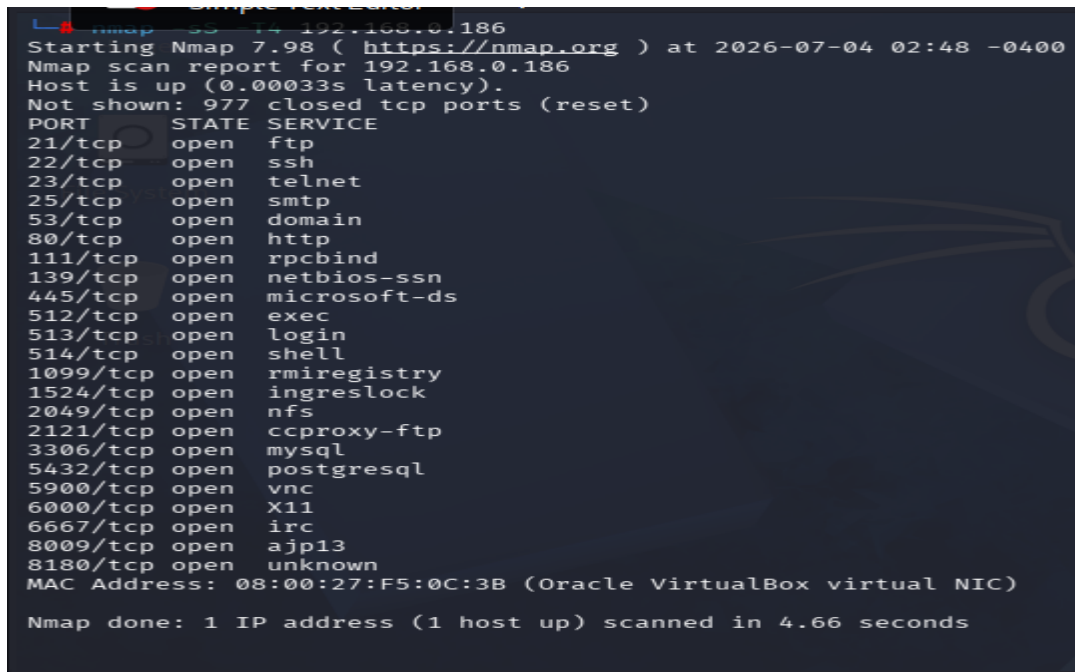
- Successfully discovered active hosts on the network.
- Identified the IP addresses of the vulnerable machines used during the assessment.

Phase 2: TCP SYN Scan

Command:- `nmap -sS -T4 192.168.0.186`

Purpose :-Performs a stealth SYN scan to identify open TCP ports while minimizing detection.

Screenshot



```
nmap -sS -T4 192.168.0.186
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-04 02:48 -0400
Nmap scan report for 192.168.0.186
Host is up (0.00033s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:F5:0C:3B (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 4.66 seconds
```

Findings

- Identified multiple open TCP ports.
- Confirmed accessible network services running on the target.

Phase 3: Service Version Detection

Command:- `nmap -sV 192.168.0.186`

Purpose:- Detects services running on open ports and determines their version numbers.

Screenshot

```
kali@kali: ~$ nmap -sV 192.168.0.186
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-04 02:49 -0400
Nmap scan report for 192.168.0.186
Host is up (0.00031s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rshcd
513/tcp   open  login?         netkit-rsh rshcd
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi        GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:F5:0C:3B (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 57.30 seconds
```

Findings

The scan identified services including:

- FTP
- SSH
- MYSQL
- HTTP
- HTTPS
- Telnet
- Postgresql

Version detection helps identify outdated software that may contain known vulnerabilities.

Phase 4: Operating System Detection

Command:- `nmap -O 192.168.0.186`

Purpose:- Attempts to identify the operating system of the target host.

Screenshot

```
(root@kali)-[/home/kali]
# nmap -O 192.168.0.186
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-04 02:53 -0400
Nmap scan report for 192.168.0.186
Host is up (0.00055s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:F5:0C:3B (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 5.98 seconds
```

Findings

Successfully estimated the operating system of the target based on TCP/IP fingerprinting.

Phase 5: Vulnerability Scanning with NSE

Command:- `nmap --script vuln 192.168.0.186`

Purpose:- Uses the Nmap Scripting Engine (NSE) to identify known vulnerabilities and security misconfigurations.

Screenshot

```

(root@kali)-[/home/kali]
└─$ nmap --script vuln 192.168.0.186
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-04 03:07 -0400
Nmap scan report for 192.168.0.186
Host is up (0.00024s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|     vsFTPD version 2.3.4 backdoor
|       State: VULNERABLE (Exploitable)
|       IDs:  BID:48539  CVE:CVE-2011-2523
|       vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|       Disclosure date: 2011-07-03
|       Exploit results:
|         Shell command: id
|         Results: uid=0(root) gid=0(root)
|       References:
|         http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|         https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|         https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|         https://www.securityfocus.com/bid/48539
|_
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
| smtp-vuln-cve2010-4344:
|_ The SMTP server is not Exim: NOT VULNERABLE
53/tcp    open  domain
80/tcp    open  http
|_ http-trace: TRACE is enabled
| http-slowloris-check:
|   VULNERABLE:
|     Slowloris DOS attack
|       State: LIKELY VULNERABLE
|       IDs:  CVE:CVE-2007-6750
|       Slowloris tries to keep many connections to the target web server open and hold
|       them open as long as possible. It accomplishes this by opening connections to
|       the target web server and sending a partial request. By doing so, it starves

```

```

|       State: LIKELY VULNERABLE
|       IDs:  CVE:CVE-2007-6750
|       Slowloris tries to keep many connections to the target web server open and hold
|       them open as long as possible. It accomplishes this by opening connections to
|       the target web server and sending a partial request. By doing so, it starves
|       the http server's resources causing Denial Of Service.
|
|       Disclosure date: 2009-09-17
|       References:
|         https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2007-6750
|         http://ha.ckers.org/slowloris/
|_ http-sql-injection:
|   Possible sql injection queries:
|     http://192.168.0.186:80/dav/?C=D%3B0%3DA%27%200R%20sqlspider
|     http://192.168.0.186:80/dav/?C=S%3B0%3DA%27%200R%20sqlspider
|     http://192.168.0.186:80/dav/?C=N%3B0%3DD%27%200R%20sqlspider
|     http://192.168.0.186:80/dav/?C=M%3B0%3DA%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=register.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=home.php&do=toggle-hints%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=user-info.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=show-log.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/?page=login.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/?page=credits.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=php-errors.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=home.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=pen-test-tool-lookup.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=documentation%2Fvulnerabilities.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=credits.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=password-generator.php%27%200R%20sqlspider&username=anonymous
|     http://192.168.0.186:80/mutillidae/index.php?page=site-footer-xss-discussion.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=text-file-viewer.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=source-viewer.php%27%200R%20sqlspider
|     http://192.168.0.186:80/mutillidae/index.php?page=installation.php%27%200R%20sqlspider

```



```
http://192.168.0.186:80/mutillidae/index.php?page=home.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=pen-test-tool-lookup.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=documentation%2Fvulnerabilities.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=credits.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=password-generator.php%27%200R%20sqlspider6username=anonymous
http://192.168.0.186:80/mutillidae/index.php?page=site-footer-xss-discussion.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=text-file-viewer.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=source-viewer.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=installation.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/?page=show-log.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=user-poll.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=notes.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=usage-instructions.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=documentation%2Fhow-to-access-Mutillidae-over-Virtual-Box-network.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=change-log.htm%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=login.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/?page=view-someones-blog.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=html5-storage.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=dns-lookup.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/?page=source-viewer.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=secret-administrative-pages.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/?page=add-to-your-blog.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=browser-info.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=capture-data.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=arbitrary-file-inclusion.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/?page=text-file-viewer.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=framing.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=view-someones-blog.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=home.php%20do-toggle-security%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=captured-data.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=add-to-your-blog.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=arbitrary-file-inclusion.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=register.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=user-info.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=show-log.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/?page=login.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/?page=credits.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=home.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=pen-test-tool-lookup.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/index.php?page=documentation%2Fvulnerabilities.php%27%200R%20sqlspider
```

```
http://192.168.0.186:80/mutillidae/index.php?page=set-background-color.php%27%200R%20sqlspider
http://192.168.0.186:80/mutillidae/?page=user-info.php%27%200R%20sqlspider
Possible sqli for forms:
  Form at path: /mutillidae/index.php, form's action: ./index.php?page=user-info.php. Fields that might be vulnerable
  - username
http-vuln-cve2017-1001000: ERROR: Script execution failed (use -d to debug)
http-csrf:
Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=192.168.0.186
Found the following possible CSRF vulnerabilities:

  Path: http://192.168.0.186:80/dvwa/
  Form id:
  Form action: login.php

  Path: http://192.168.0.186:80/dvwa/login.php
  Form id:
  Form action: login.php

  Path: http://192.168.0.186:80/twiki/TWikiDocumentation.html
  Form id:
  Form action: http://TWiki.org/cgi-bin/passwd/TWiki/WebHome

  Path: http://192.168.0.186:80/twiki/TWikiDocumentation.html
  Form id:
  Form action: http://TWiki.org/cgi-bin/passwd/Main/WebHome

  Path: http://192.168.0.186:80/twiki/TWikiDocumentation.html
  Form id:
  Form action: http://TWiki.org/cgi-bin/edit/TWiki/

  Path: http://192.168.0.186:80/twiki/TWikiDocumentation.html
  Form id:
  Form action: http://TWiki.org/cgi-bin/view/TWiki/TWikiSkins

  Path: http://192.168.0.186:80/twiki/TWikiDocumentation.html
  Form id:
  Form action: http://TWiki.org/cgi-bin/manage/TWiki/ManagingWebs

  Path: http://192.168.0.186:80/mutillidae/index.php?page=register.php
  Form id: id-bad-cred-tr
  Form action: index.php?page=register.php
```

```

| Form id: id-bad-cred-tr
| Form action: index.php?page=register.php
|
| Path: http://192.168.0.186:80/mutillidae/index.php?page=user-info.php
| Form id: id-bad-cred-tr
| Form action: ../index.php?page=user-info.php
| http-enum:
| /tikiwiki/: Tikiwiki
| /test/: Test page
| /phpinfo.php: Possible information file
| /phpMyAdmin/: phpMyAdmin
| /doc/: Potentially interesting directory w/ listing on 'apache/2.2.8 (ubuntu) dav/2'
| /icons/: Potentially interesting folder w/ directory listing
| /index/: Potentially interesting folder
| http-stored-xss: Couldn't find any stored XSS vulnerabilities.
| http-fileupload-exploiter:
|
| Couldn't find a file-type field.
| http-dombased-xss: Couldn't find any DOM based XSS.
| 111/tcp open  rpcbind
| 139/tcp open  netbios-ssn
| 445/tcp open  microsoft-ds
| 512/tcp open  exec
| 513/tcp open  login
| 514/tcp open  shell
| 1099/tcp open rmiregistry
| rmi-vuln-classloader:
| VULNERABLE:
|   RMI registry default configuration remote code execution vulnerability
|   State: VULNERABLE
|     Default configuration of RMI registry allows loading classes from remote URLs which can lead to remote code execution.
|
| References:
|   https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/multi/misc/java_rmi_server.rb
| 1524/tcp open  ingreslock
| 2049/tcp open  nfs
| 2121/tcp open  ccproxy-ftp
| 3306/tcp open  mysql
| 5432/tcp open  postgresql
| ssl-dh-params:
| VULNERABLE:

```

```

| JSESSIONID:
| httponly flag not set
| http-enum:
| /admin/: Possible admin folder
| /admin/index.html: Possible admin folder
| /admin/login.html: Possible admin folder
| /admin/admin.html: Possible admin folder
| /admin/account.html: Possible admin folder
| /admin/admin_login.html: Possible admin folder
| /admin/home.html: Possible admin folder
| /admin/admin-login.html: Possible admin folder
| /admin/adminLogin.html: Possible admin folder
| /admin/controlpanel.html: Possible admin folder
| /admin/cp.html: Possible admin folder
| /admin/index.jsp: Possible admin folder
| /admin/login.jsp: Possible admin folder
| /admin/admin.jsp: Possible admin folder
| /admin/home.jsp: Possible admin folder
| /admin/controlpanel.jsp: Possible admin folder
| /admin/admin-login.jsp: Possible admin folder
| /admin/cp.jsp: Possible admin folder
| /admin/account.jsp: Possible admin folder
| /admin/admin_login.jsp: Possible admin folder
| /admin/adminLogin.jsp: Possible admin folder
| /manager/html/upload: Apache Tomcat (401 Unauthorized)
| /manager/html: Apache Tomcat (401 Unauthorized)
| /admin/view/javascript/fckeditor/editor/filemanager/connectors/test.html: OpenCart/FCKeditor File upload
| /admin/includes/FCKeditor/editor/filemanager/upload/test.html: ASP Simple Blog / FCKeditor File Upload
| /admin/jscript/upload.html: Lizard Cart/Remote File upload
| /webdav/: Potentially interesting folder
| MAC Address: 08:00:27:F5:0C:3B (Oracle VirtualBox virtual NIC)
|
| Host script results:
| _smb-vuln-regsvc-dos: ERROR: Script execution failed (use -d to debug)
| _smb-vuln-ms10-054: false
| _smb-vuln-ms10-061: false
|
| Nmap done: 1 IP address (1 host up) scanned in 329.76 seconds

```

Findings

The vulnerability scan identified several security issues including:

- VSFTPD Backdoor
- RMI-Vuln-Classloader
- HTTP Slowloris Vulnerability

Results Summary

The assessment successfully identified:

- Active hosts on the network
- Open TCP ports
- Running network services
- Service versions
- Operating system information
- Potential security vulnerabilities
- Hidden web directories and applications

The lab demonstrates how Nmap can be used throughout the reconnaissance and enumeration phases of a penetration test.
